Foundation Models for Satellite Earth Observation: A Survey

Satellite Earth observation is entering a foundation-model era in which large pretrained models learn geospatial representations from multispectral, hyperspectral, SAR, optical, temporal, and multimodal archives and are adapted to land-cover mapping, segmentation, object detection, change detection, disaster response, agriculture, climate monitoring, and other downstream Earth-science tasks. This survey synthesizes an OpenAlex-derived academic corpus of 300 papers collected during this run and organizes the literature around remote-sensing pretraining, sensor fusion, vision-language alignment, semantic understanding, temporal modeling, datasets, benchmarks, adaptation, robustness, physical grounding, and operational deployment. The survey argues that foundation models for Earth observation must be evaluated not only by average task accuracy but also by geographic transfer, sensor shift, temporal generalization, uncertainty, label efficiency, and scientific utility under real observing constraints.

Keywords:

foundation models; satellite Earth observation; remote sensing; geospatial AI; self-supervised learning; multimodal learning; vision-language models; transfer learning

1 Introduction

Satellite Earth observation provides repeated, global, multi-sensor measurements of the planet. Optical, multispectral, hyperspectral, thermal, radar, lidar, and derived geospatial products support land-cover mapping, crop monitoring, urban analysis, disaster response, ecosystem monitoring, carbon accounting, climate adaptation, and security-relevant applications. The scale and diversity of these archives make them a natural substrate for foundation models: large pretrained systems that learn reusable geospatial representations and can be adapted to many downstream tasks [20, 54, 83, 108, 151, 182, 194, 201].

Foundation models are appealing in Earth observation because labeled data are expensive, unevenly distributed, and often task-specific. A field boundary dataset in one country may not transfer to another; a flood map may depend on sensor availability and cloud cover; a land-cover label may be ambiguous across seasons. Pretraining on large unlabeled satellite archives can reduce annotation needs and improve transfer, but only if the model captures the properties that make Earth observation distinct: geolocation, scale, spectral bands, temporal dynamics, clouds, sensor physics, and domain shift [20, 54, 83, 108, 151, 182, 194, 201].

This survey treats satellite foundation models as geospatial systems rather than generic image encoders. Their success depends on data curation, sensor harmonization, coordinate-aware representations, temporal sampling, adaptation protocols, and evaluation across regions, sensors, seasons, and tasks. We therefore organize the literature around pretraining, sensor modality, vision-language alignment, downstream tasks, temporal modeling, benchmarks, transfer, robustness, and deployment.

2 Background

2.1 Satellite Earth Observation Data

Earth observation data differ from ordinary natural images. Pixels are georeferenced, bands have physical meanings, resolutions vary from sub-meter imagery to kilometer-scale products, revisit intervals differ by platform, and cloud, atmosphere, terrain, sun angle, speckle, and orbital geometry influence measurements. Optical missions such as Sentinel-2 and Landsat provide multispectral time series; SAR missions such as Sentinel-1 provide all-weather backscatter; other missions supply thermal, hyperspectral, lidar, or derived products. Foundation models must respect this heterogeneity [17, 20, 24, 64, 83, 108, 182, 201].

2.2 What Makes a Foundation Model in Remote Sensing

In this survey, a remote-sensing foundation model is not merely a large classifier. It is a pretrained representation model or generative model designed to transfer across geographies, sensors, resolutions, times, or tasks. Pretraining may use masked reconstruction, contrastive learning, image-text alignment, temporal prediction, multi-sensor fusion, geolocation-aware objectives, or generative modeling. Adaptation may involve linear probing, fine-tuning, adapters, prompting, retrieval, or zero-shot inference [20, 54, 83, 108, 151, 182, 194, 201].

2.3 From Remote Sensing Pipelines to General Geospatial AI

Traditional remote-sensing pipelines were often task-specific: collect labels, train a model for a region and sensor, validate on a held-out split, and deploy cautiously. Foundation models promise a more reusable stack: one backbone can support land-cover mapping, segmentation, detection, change detection, crop classification, disaster mapping, and captioning. The challenge is that geospatial transfer is harder than benchmark transfer because label taxonomies, resolution, seasonality, sensor calibration, and regional ecology all change.

3 Methods

3.1 Search Protocol and Corpus Construction

The literature corpus for this survey was generated during this run from the OpenAlex Works API after the arXiv API returned HTTP 429 during task028. Queries combined foundation model, pretrained model, self-supervised learning, masked autoencoder, contrastive learning, transformer, vision-language model, CLIP, large language model, satellite imagery, Earth observation, remote sensing, geospatial AI, multispectral, hyperspectral, SAR, Sentinel, Landsat, semantic segmentation, object detection, change detection, land cover, time series, multimodal fusion, zero-shot, few-shot, transfer learning, domain adaptation, datasets, benchmarks, and representative model or dataset names such as SatMAE, Scale-MAE, RingMo, Prithvi, TerraMind, Clay, Galileo, SkySense, SpectralGPT, RemoteCLIP, GeoCLIP, BigEarthNet, EuroSAT, SEN12MS, xView, DOTA, and FAIR1M. Records were retained when title or abstract evidence explicitly connected satellite or remote-sensing Earth observation with foundation-style pretraining, transfer, multimodal learning, large models, or foundation-model downstream adaptation. The final bibliography cites 300 unique scholarly works; arXiv eprints are mapped in ref.json when available.

3.2 Taxonomy

We organize the field along eight axes. The first is *sensor modality*: optical RGB, multispectral, hyperspectral, SAR, thermal, lidar, elevation, weather or climate products, and fused multimodal stacks. The second is *pretraining signal*: masked reconstruction, contrastive learning, temporal prediction, cross-sensor alignment, image-text alignment, generative modeling, or supervised multi-task pretraining. The third is *spatial scale*: patch, object, scene, region, country, and global grids. The fourth is *temporal structure*: single-date imagery, seasonal time series, event-driven pairs, and long-term monitoring. The fifth is *adaptation*: frozen features, linear probes, full fine-tuning, adapters, prompts, retrieval, and zero-shot inference. The sixth is *task*: classification, segmentation, detection, change detection, retrieval, captioning, VQA, and scientific regression. The seventh is *evaluation*: in-domain accuracy, cross-region transfer, sensor transfer, temporal holdout, label efficiency, uncertainty, and calibration. The eighth is *deployment*: cloud coverage, latency, memory, licensing, reproducibility, scientific validity, and operational risk.

3.3 Remote-Sensing Foundation Models and Self-Supervised Pretraining

Self-supervised pretraining is the dominant route to foundation models in Earth observation. Masked autoencoding exploits spatial and spectral redundancy; contrastive learning aligns augmentations, locations, dates, or modalities; temporal pretraining learns seasonal and phenological structure; multi-task pretraining can bind heterogeneous labels. Strong models must avoid trivial shortcuts such as memorizing geography while still exploiting location and time as meaningful context [20, 54, 83, 108, 151, 182, 194, 201].

3.4 Multispectral, Hyperspectral, SAR, and Multimodal Sensing

Sensor heterogeneity is both a strength and a difficulty. Multispectral data encode vegetation, water, soil, and urban materials beyond RGB. Hyperspectral data offer fine spectral discrimination but are less widely available. SAR provides cloud-penetrating structure and moisture sensitivity but includes speckle and geometry effects. Multimodal foundation models can align these sources, but alignment requires careful handling of resolution, noise, temporal mismatch, and missing modalities [17, 20, 24, 64, 83, 108, 182, 201].

3.5 Vision-Language and Geospatial Language Models

Vision-language models connect satellite imagery to captions, tags, reports, maps, and natural-language queries. CLIP- style alignment enables zero-shot classification and retrieval; geospatial language models can support question answering and human-in-the-loop analysis; multimodal large language models can combine visual evidence with task instructions. The risk is weak grounding: text scraped from the web or map metadata may be noisy, biased, or spatially misaligned [151, 168, 182, 191, 233, 251, 255, 256].

3.6 Semantic Segmentation, Detection, and Scene Understanding

Many operational Earth-observation tasks are dense or object-centered. Semantic segmentation maps land cover, crops, water, roads, buildings, burn scars, or clouds. Object detection finds ships, vehicles, aircraft, buildings, or infrastructure. Scene classification summarizes land use or environmental context. Foundation models can reduce labels for these tasks, but they must handle class imbalance, tiny objects, resolution changes, and ambiguous boundaries [20, 24, 54, 83, 108, 182, 194, 201].

3.7 Change Detection, Time Series, and Spatiotemporal Modeling

Earth observation is intrinsically temporal. Change-detection models compare images across dates to identify disasters, construction, deforestation, flooding, mining, or seasonal shifts. Time-series foundation models can learn phenology, land management, and climate variability. Spatiotemporal models must distinguish meaningful change from clouds, shadows, sensor differences, missing acquisitions, and registration errors [17, 62, 64, 108, 182, 194, 201, 211].

3.8 Datasets, Benchmarks, and Evaluation Protocols

Benchmarks have shaped remote-sensing machine learning, but foundation models expose their limitations. Random patch splits may overestimate transfer because nearby samples share geography. Small label sets may hide class-taxonomy differences. A useful benchmark suite should include cross-region, cross-sensor, cross-season, low-label, and out-of-distribution protocols, with transparent metadata and task definitions [20, 54, 83, 108, 151, 182, 194, 201].

3.9 Adaptation, Efficient Fine-Tuning, and Transfer Learning

Foundation models are valuable only when they adapt efficiently. Linear probes test representation quality; full fine-tuning maximizes task performance but can overfit; adapters and LoRA reduce parameter cost; prompts and retrieval enable lightweight control; zero-shot and few-shot protocols test whether semantics transfer without dense labels. The field needs clearer reporting of data budgets, hyperparameters, and target-domain leakage [24, 26, 83, 108, 151, 164, 182, 194].

3.10 Robustness, Physical Grounding, and Operational Earth Science

Operational Earth science requires more than benchmark accuracy. Models should remain reliable under clouds, haze, seasonality, sensor upgrades, geographic domain shift, land-cover change, disasters, and climate trends. They should report uncertainty, respect physical constraints where possible, and expose provenance. Scientific users need models that support measurement and inference, not only visually plausible predictions [20, 24, 108, 151, 164, 182, 194, 201].

3.11 Representative Literature Map

The following map cites the retained corpus by theme, showing how the literature spans foundation-model pretraining, multispectral and SAR sensing, vision-language alignment, segmentation and detection, temporal analysis, benchmarks, adaptation, robustness, and operational Earth science.

Remote-sensing foundation models and self-supervised pretraining The remote-sensing foundation models and self-supervised pretraining cluster contains work that develops methods, systems, datasets, or analyses central to this survey [17, 24, 54, 64, 83, 151, 182, 194, 201]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [4, 26, 51, 110, 165, 170, 191, 197, 241]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [59, 95, 111, 118, 119, 181, 199, 252, 274]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [23, 97, 115, 124, 161, 223, 224, 244, 250]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [12, 35, 79, 89, 179, 206, 210, 226, 231]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [7, 42, 53, 105, 134, 207, 215, 253, 289]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [10, 57, 78, 129, 138, 155, 218, 246, 282]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [34, 67, 92, 112, 122, 133, 258, 265, 275]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [31, 40, 88, 94, 130, 159, 254, 270, 281]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [3, 22, 41, 45, 96, 106, 209, 236, 257]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [37, 44, 49, 90, 99, 121, 136, 220, 268]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [38, 58, 66, 73, 131, 184, 262, 276, 298]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [46, 144, 189].

Multispectral, hyperspectral, sar, and multimodal sensing The multispectral, hyperspectral, SAR, and multimodal sensing cluster contains work that develops methods, systems, datasets, or analyses central to this survey [16, 20, 29, 36, 50, 80, 85, 104, 264]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [61, 68, 84, 107, 140, 146, 153, 269, 293]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [39, 93, 117, 123, 176, 217, 228, 261].

Vision-language and geospatial language models The vision-language and geospatial language models cluster contains work that develops methods, systems, datasets, or analyses central to this survey [48, 91, 120, 145, 168, 233, 242, 256, 297]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [205, 237, 245, 291].

Semantic segmentation, detection, and scene understanding The semantic segmentation, detection, and scene understanding cluster contains work that develops methods, systems, datasets, or analyses central to this survey [13, 60, 74, 98, 178, 225, 279, 284, 294]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [33, 76, 103, 135, 142, 152, 160, 235, 283]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [5, 15, 162, 163, 183, 222, 234, 287].

Change detection, time series, and spatiotemporal modeling The change detection, time series, and spatiotemporal modeling cluster contains work that develops methods, systems, datasets, or analyses central to this survey [6, 62, 81, 108, 185, 211, 227, 263, 295]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [32, 52, 69, 116, 216, 232, 266, 272, 299]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [200, 277].

Datasets, benchmarks, and evaluation protocols The datasets, benchmarks, and evaluation protocols cluster contains work that develops methods, systems, datasets, or analyses central to this survey [19, 25, 27, 164, 204, 208, 240, 251, 255]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [65, 71, 113, 148, 196, 280, 286, 296, 300]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [9, 28, 47, 77, 132, 172, 174, 249, 267]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [14, 147, 158, 171, 173, 177, 271, 278, 288]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [21, 100, 156, 180, 188, 193, 239, 259, 290]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [1, 143, 149, 195, 202, 203, 229, 230, 247]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [43, 82, 86, 109, 157, 213, 214, 221, 248]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [30, 101, 150, 169, 212, 219, 238, 260, 273]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [2, 63, 75, 127, 128, 137, 186, 243, 292]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [18, 56, 70, 102, 126, 175, 190, 192, 198]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [8, 11, 125, 141, 154, 187].

Adaptation, efficient fine-tuning, and transfer learning The adaptation, efficient fine-tuning, and transfer learning cluster contains work that develops methods, systems, datasets, or analyses central to this survey [166, 167].

Robustness, physical grounding, and operational earth science The robustness, physical grounding, and operational Earth science cluster contains work that develops methods, systems, datasets, or analyses central to this survey [55, 72, 87, 114, 139, 285].

4 Challenges and Prospects

4.1 Geographic and Temporal Generalization

Earth-observation models often perform well near their training distribution and degrade across continents, climates, seasons, or years. Future benchmarks should emphasize spatial and temporal holdouts, not only random patch splits.

4.2 Sensor-Aware Multimodal Learning

A foundation model should know which bands, resolutions, noise processes, and acquisition conditions generated its input. Sensor-aware tokenization, missing-modality handling, and physically meaningful fusion are central open problems.

4.3 Label Semantics and Task Taxonomies

Remote-sensing labels are not universal. Land-cover classes, crop labels, disaster masks, and object definitions differ across datasets and agencies. Foundation models need mechanisms for mapping, explaining, and adapting label semantics.

4.4 Trust, Uncertainty, and Scientific Validity

Operational users need calibrated uncertainty, provenance, and failure warnings. For scientific use, a model must not only classify images but also support credible measurement, change attribution, and decision making under uncertainty.

4.5 Open, Reproducible Geospatial Foundation Models

The field will benefit from open datasets, transparent pretraining recipes, documented licensing, energy and carbon accounting, reproducible evaluation suites, and model cards that describe geographic coverage, sensor support, and known failure modes.

5 Conclusion

Foundation models for satellite Earth observation promise reusable representations for geospatial AI, but their value depends on respecting the structure of Earth-observation data. The strongest systems will integrate multispectral, SAR, temporal, language, and physical context; adapt efficiently to downstream tasks; and demonstrate transfer across regions, seasons, sensors, and hazards. The next phase of the field should move from single-benchmark gains toward trustworthy, sensor-aware, scientifically grounded models for operational Earth observation.

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